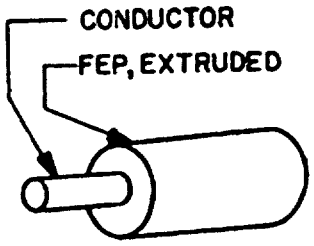
MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, FLUORINATED ETHYLENE PROPYLENE (FEP) INSULATED, 200°C, 600 VOLTS

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the wire described herein shall consist of this document and the latest issue of MIL-W-16878.



SH11879

FIGURE 1. Construction.

TABLE I. Construction details.

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire	
					Diameter (inch)	
		Material 2/	Coating		Min	Max
32	1 X 32	Copper	Silver	0.0080	0.024	0.032
32	1 X 32	C.C. Steel	Silver	.0080	.024	.032
32	7 X 40	Copper	Silver	.010	.026	.034
30	1 X 30	Copper	Silver	.0100	.026	.034
30	1 X 30	C.C. Steel	Silver	.0100	.026	.034
30	7 X 38	Copper	Silver	.012	.028	.036

See footnotes at end of table

*U.S. GOVERNMENT PRINTING OFFICE: 1981—703-023/5789

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:28Z
Check the source to verify that this is the current version before use.

TABLE I. Construction details.- Continued

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire Diameter (inch)	
		Material 2/	Coating		Min	Max
28	1 X 32	Copper	Silver	0.0126	0.029	0.037
28	1 X 38	C.C. Steel	Silver	.0126	.029	.037
28	7 X 36	Copper	Silver	.015	.031	.039
26	1 X 26	Copper	Silver	.0159	.032	.040
26	1 X 26	C.C. Steel	Silver	.0159	.032	.040
26	7 X 34	Copper	Silver	.019	.035	.043
26	19 X 38	Copper	Silver	.020	.035	.043
1/ 24	1 X 24	Copper	Silver	.0201	.036	.044
1/ 24	1 X 24	C.C. Steel	Silver	.0201	.036	.044
24	7 X 32	Copper	Silver	.024	.040	.048
24	19 X 36	Copper	Silver	.025	.040	.048
1/ 22	1 X 22	Copper	Silver	.0254	.041	.050
1/ 22	1 X 22	C.C. Steel	Silver	.0254	.041	.050
22	7 X 30	Copper	Silver	.030	.046	.054
22	19 X 34	Copper	Silver	.032	.046	.054
20	1 X 20	Copper	Silver	.0320	.058	.066
20	7 X 28	Copper	Silver	.038	.064	.072
20	19 X 32	Copper	Silver	.040	.064	.072
18	1 X 18	Copper	Silver	.0403	.056	.066
18	7 X 26	Copper	Silver	.048	.064	.074
18	19 X 30	Copper	Silver	.050	.064	.074
16	1 X 16	Copper	Silver	.0508	.067	.081
16	19 X 29	Copper	Silver	.057	.073	.087
14	1 X 14	Copper	Silver	.0641	.080	.094
14	19 X 27	Copper	Silver	.072	.088	.102
12	19 X 25	Copper	Silver	.091	.107	.121
10	37 X 26	Copper	Silver	.111	.127	.141
8	133 X 29	Copper	Silver	.169	.185	.199

1/ Inactive for new design.
2/ C.C. Steel stands for copper clad steel.

REQUIREMENTS:

Visual and mechanical inspection: Applicable.
Spark test: 3.4 kilovolts (kV).
Impulse dielectric test: 6.5 kV.
Tank test: 2.0 kV.
Dielectric withstanding voltage: 2.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20°C

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:28Z
Check the source to verify that this is the current version before use.

K = 50,000
D = Maximum average diameter of finished wire
d = Conductor diameter

Cold bend: Condition 4 hours at minus 65°C ± 1°C (see table II).

TABLE II. Cold bend mandrel sizes.

Wire size	Cold bend mandrel (maximum diameter) (inches)
32 through 16	1
14 through 10	2
8	3

Concentricity: 70 percent (minimum).
Surface resistance: 5 megohms (minimum).
Wrap back: Not applicable.
Heat resistance: Condition at 230°C.
Flammability: Not applicable.
Soldering: Applicable.
Heat aging: Not applicable.
Insulation tensile strength: 2000 pounds force per square inch (minimum).
Insulation elongation: 100 percent (minimum).
Stripe durability: Applicable.
Fungus resistance: Applicable.
Maximum direct current resistance of finished wire: See table I of MIL-W-16878.

Military part number (see MIL-W-16878):

M16878/ 11 B C B 905

Military designation — M16878/ 11
Specification sheet — 11

B: Conductor material
C: Conductor size
B: Conductor stranding
905: Color code

Review activities:
Navy - AS, EC

User activities:
Navy - OS, MC

Preparing activity:
Navy - SH
(Project 6145-N300-10)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:28Z
Check the source to verify that this is the current version before use.